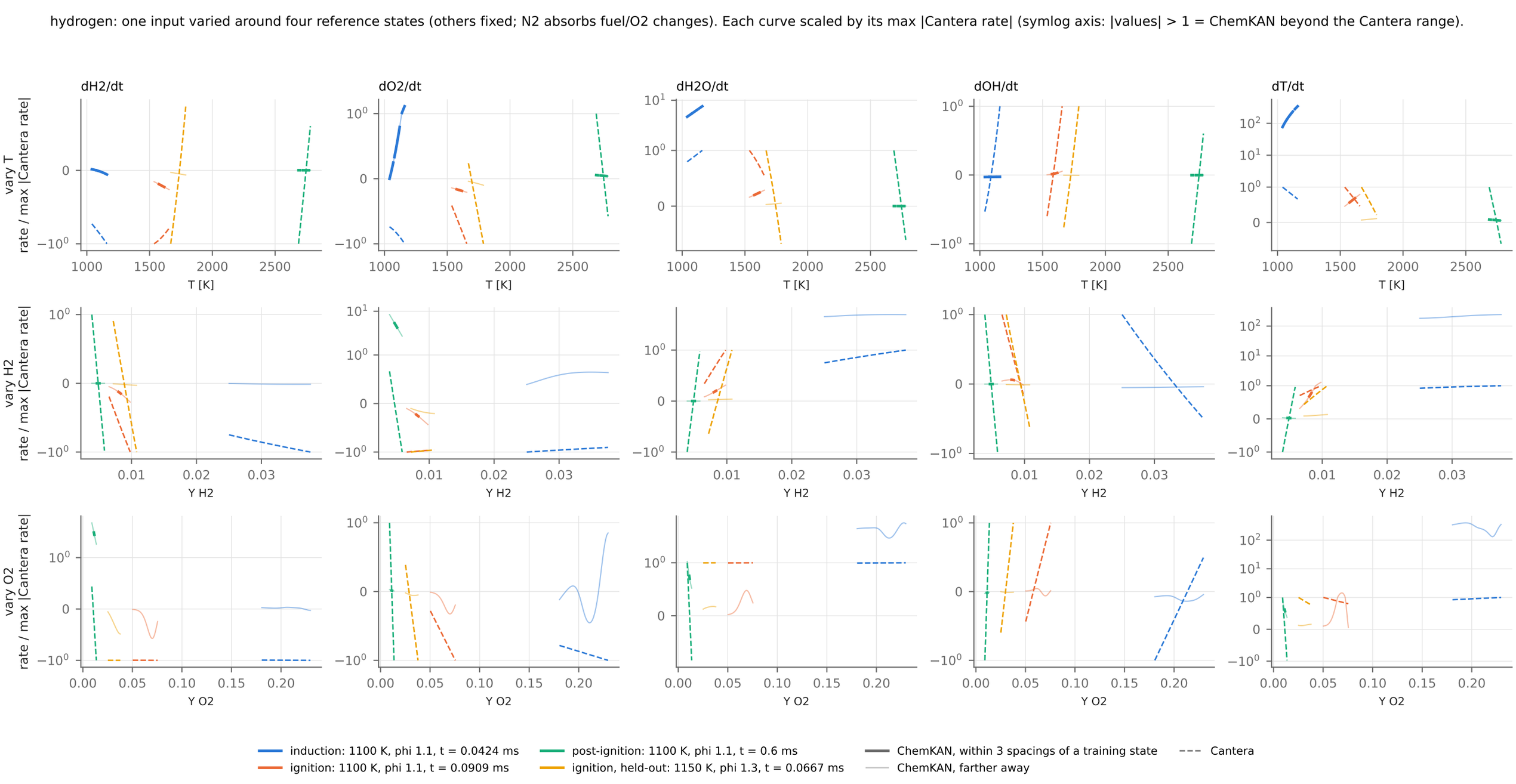

hydrogen: one input varied around four reference states (others fixed; N2 absorbs fuel/O2 changes). Each curve scaled by its max |Cantera rate| (symlog axis: |values| > 1 = ChemKAN beyond the Cantera range).
dH2/dt
dO2/dt
dH2O/dt
dOH/dt
dT/dt
vary T
vary H2
vary O2
rate / max |Cantera rate|
T [K]
Y H2
Y O2
induction: 1100 K, phi 1.1, t = 0.0424 ms
ignition: 1100 K, phi 1.1, t = 0.0909 ms
post-ignition: 1100 K, phi 1.1, t = 0.6 ms
ignition, held-out: 1150 K, phi 1.3, t = 0.0667 ms
ChemKAN, within 3 spacings of a training state
ChemKAN, farther away
Cantera